

NIRMIQ RAG Reference Notes

Hybrid retrieval combines lexical BM25 search with optional semantic vector search. BM25 is strong for exact academic terms, formulas, and named concepts. Vector search is useful when the student uses paraphrased wording. Reciprocal Rank Fusion combines ranked lists without requiring score calibration. NIRMIQ uses this flow to keep retrieval useful even when one retriever misses a passage.

Citation-aware answers must point back to source chunks. A strong answer should include page references, evidence snippets, and enough context for the student to verify the claim.

Hallucination Control

NIRMIQ reduces hallucinations by grounding answers in retrieved chunks, checking citation coverage, and rewriting unsupported generated claims into source-only evidence. If the retrieved context is weak or unrelated, the assistant should abstain instead of guessing. This is especially important for exam preparation and research-paper drafting.

A low-memory local profile should prefer smaller quantized models, bounded context windows, bounded prediction length, and short model keep-alive settings. Retrieval quality is prioritized over model size.

Evaluation Metrics

Retrieval quality can be measured with Recall at K, Mean Reciprocal Rank, normalized Discounted Cumulative Gain, and citation accuracy. Recall at K checks whether expected evidence appears in the top retrieved chunks. MRR rewards placing the first correct source earlier. nDCG rewards ranking multiple relevant chunks near the top. Citation accuracy checks whether the final cited sources support the answer.